

Initial exploration

EXPLORATORY DATA ANALYSIS IN PYTHON



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Exploratory Data Analysis

The process of reviewing and cleaning data to...

- derive insights
- generate hypotheses



A first look with .head()

```
books = pd.read_csv("books.csv")
books.head()
```

name	author	rating	year	genre
10-Day Green Smoothie Cleanse	JJ Smith	4.73	2016	Non Fiction
11/22/63: A Novel	Stephen King	4.62	2011	Fiction
12 Rules for Life	Jordan B. Peterson	4.69	2018	Non Fiction
1984 (Signet Classics)	George Orwell	4.73	2017	Fiction
5,000 Awesome Facts	National Geographic Kids	4.81	2019	Childrens

Gathering more .info()

```
books.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 350 entries, 0 to 349
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
--  --
 0   name        350 non-null    object
 1   author      350 non-null    object
 2   rating      350 non-null    float64
 3   year        350 non-null    int64
 4   genre       350 non-null    object
dtypes: float64(1), int64(1), object(3)
memory usage: 13.8+ KB
```

A closer look at categorical columns

```
books.value_counts("genre")
```

```
genre
Non Fiction    179
Fiction        131
Childrens      40
dtype: int64
```

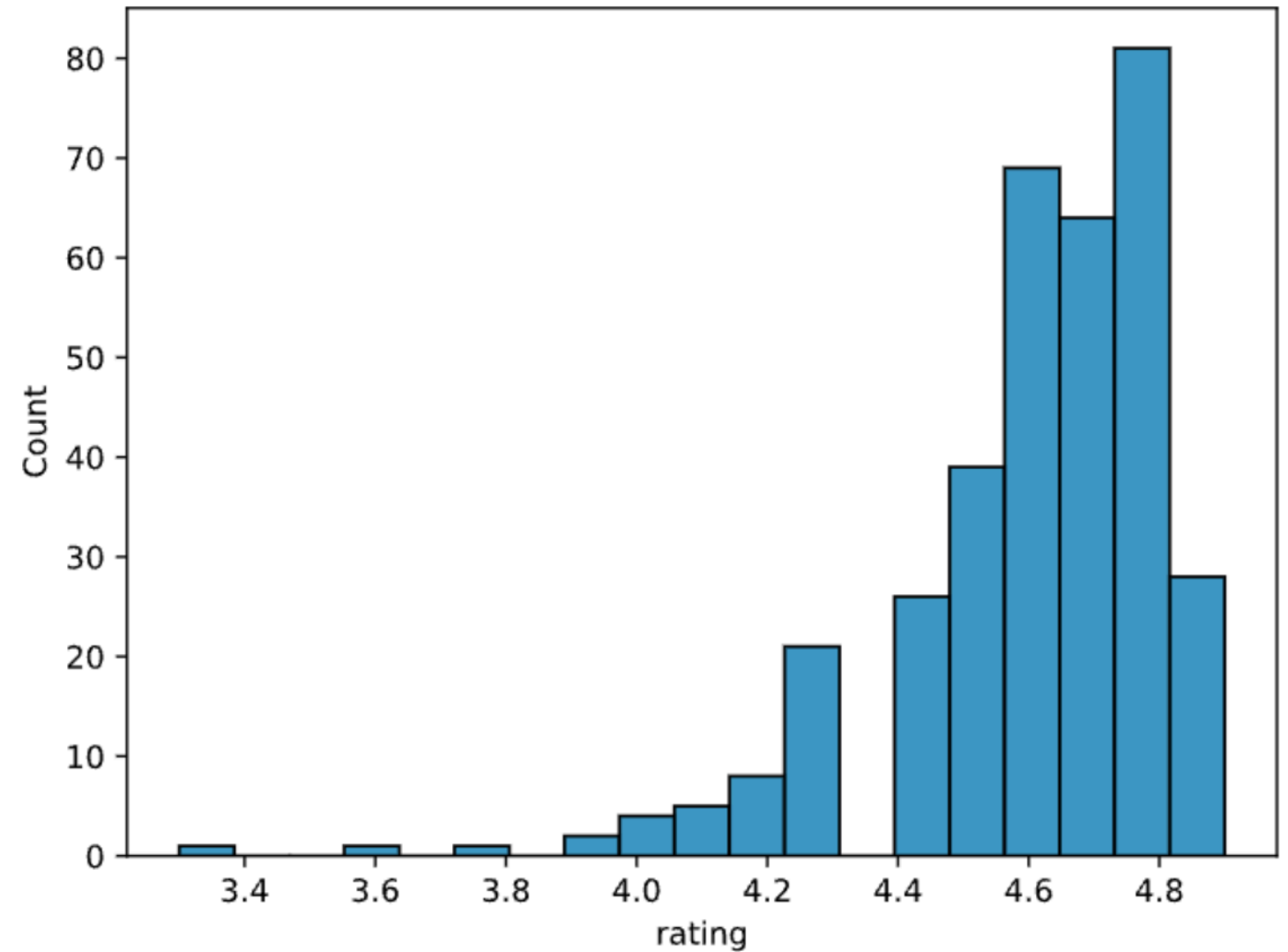
.describe() numerical columns

```
books.describe()
```

	rating	year
count	350.000000	350.000000
mean	4.608571	2013.508571
std	0.226941	3.284711
min	3.300000	2009.000000
25%	4.500000	2010.000000
50%	4.600000	2013.000000
75%	4.800000	2016.000000
max	4.900000	2019.000000

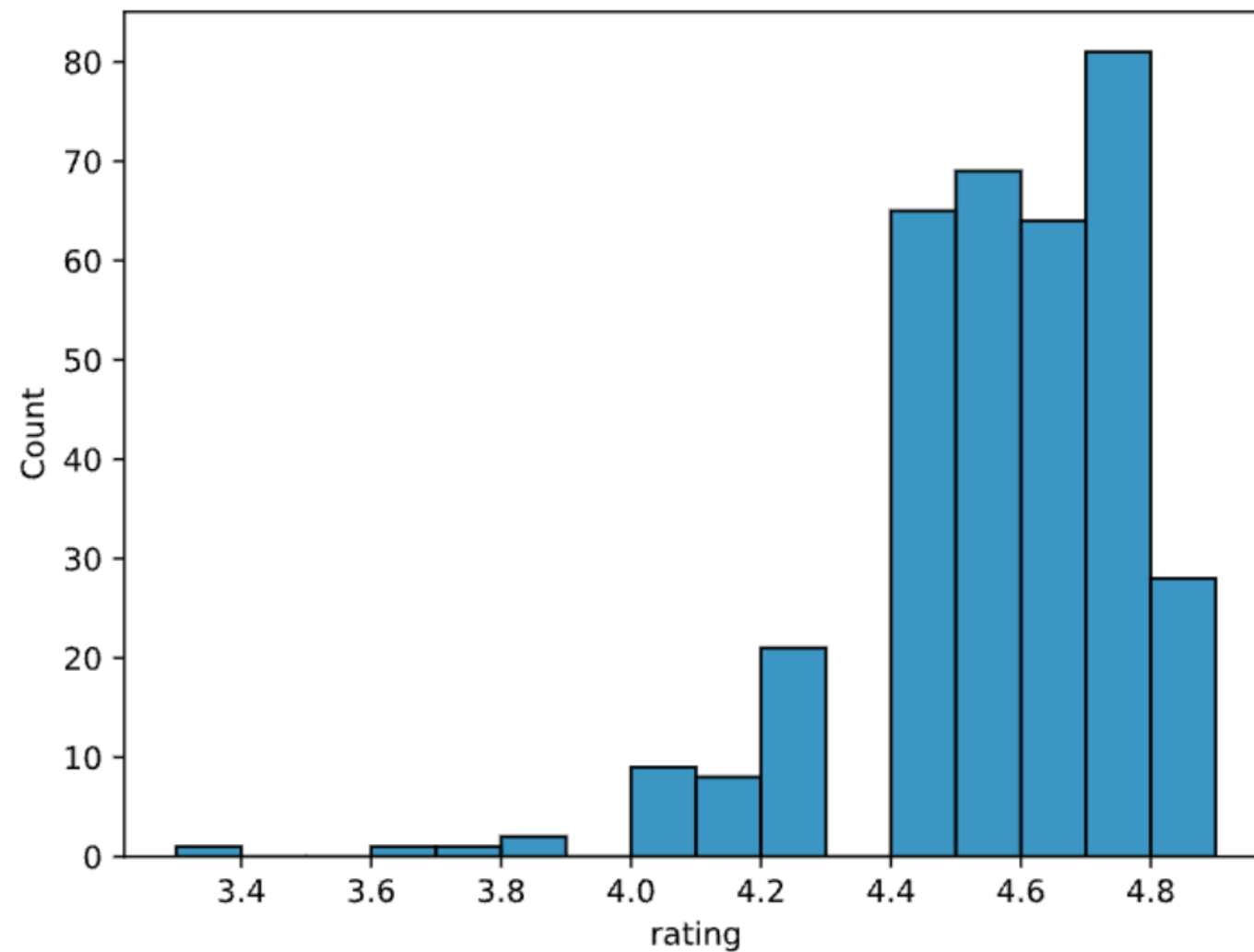
Visualizing numerical data

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.histplot(data=books, x="rating")
plt.show()
```



Adjusting bin width

```
sns.histplot(data=books, x="rating", binwidth=.1)  
plt.show()
```



Let's practice!

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Data validation

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Validating data types

```
books.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 350 entries, 0 to 349
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
--  --
 0   name        350 non-null    object
 1   author      350 non-null    object
 2   rating      350 non-null    float64
 3   year        350 non-null    float64
 4   genre       350 non-null    object
dtypes: float64(1), int64(1), object(3)
memory usage: 13.8+ KB
```

```
books.dtypes
```

```
name        object
author      object
rating       float64
year         float64
genre        object
dtype: object
```

Updating data types

```
books["year"] = books["year"].astype(int)
books.dtypes
```

```
name      object
author    object
rating    float64
year      int64
genre     object
dtype: object
```

Updating data types

Type	Python Name
String	<code>str</code>
Integer	<code>int</code>
Float	<code>float</code>
Dictionary	<code>dict</code>
List	<code>list</code>
Boolean	<code>bool</code>

Validating categorical data

```
books["genre"].isin(["Fiction", "Non Fiction"])
```

```
0      True
1      True
2      True
3      True
4     False
...
345    True
346    True
347    True
348    True
349    False
Name: genre, Length: 350, dtype: bool
```

Validating categorical data

```
~books["genre"].isin(["Fiction", "Non Fiction"])
```

```
0      False
1      False
2      False
3      False
4       True
...
345     False
346     False
347     False
348     False
349      True
Name: genre, Length: 350, dtype: bool
```


Validating categorical data

```
books[books["genre"].isin(["Fiction", "Non Fiction"])]
```

```
books[books["genre"].isin(["Fiction", "Non Fiction"])]
```

	name	author	rating	year	genre
0	10-Day Green Smoothie Cleanse	JJ Smith	4.7	2016	Non Fiction
1	11/22/63: A Novel	Stephen King	4.6	2011	Fiction
2	12 Rules for Life	Jordan B. Peterson	4.7	2018	Non Fiction
3	1984 (Signet Classics)	George Orwell	4.7	2017	Fiction
5	A Dance with Dragons	George R. R. Martin	4.4	2011	Fiction

Validating numerical data

```
books.select_dtypes("number").head()
```

	rating	year
0	4.7	2016
1	4.6	2011
2	4.7	2018
3	4.7	2017
4	4.8	2019

Validating numerical data

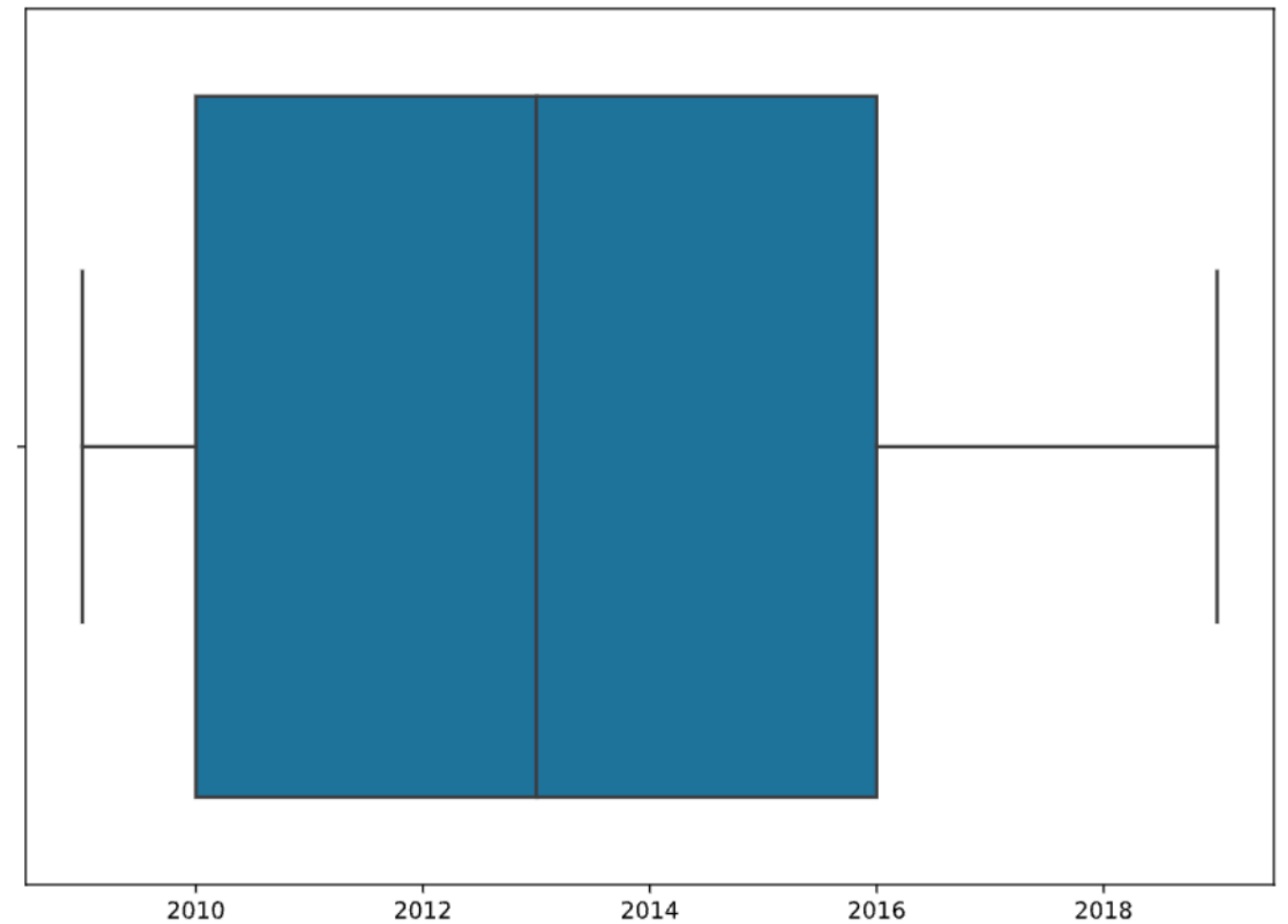
```
books["year"].min()
```

2009

```
books["year"].max()
```

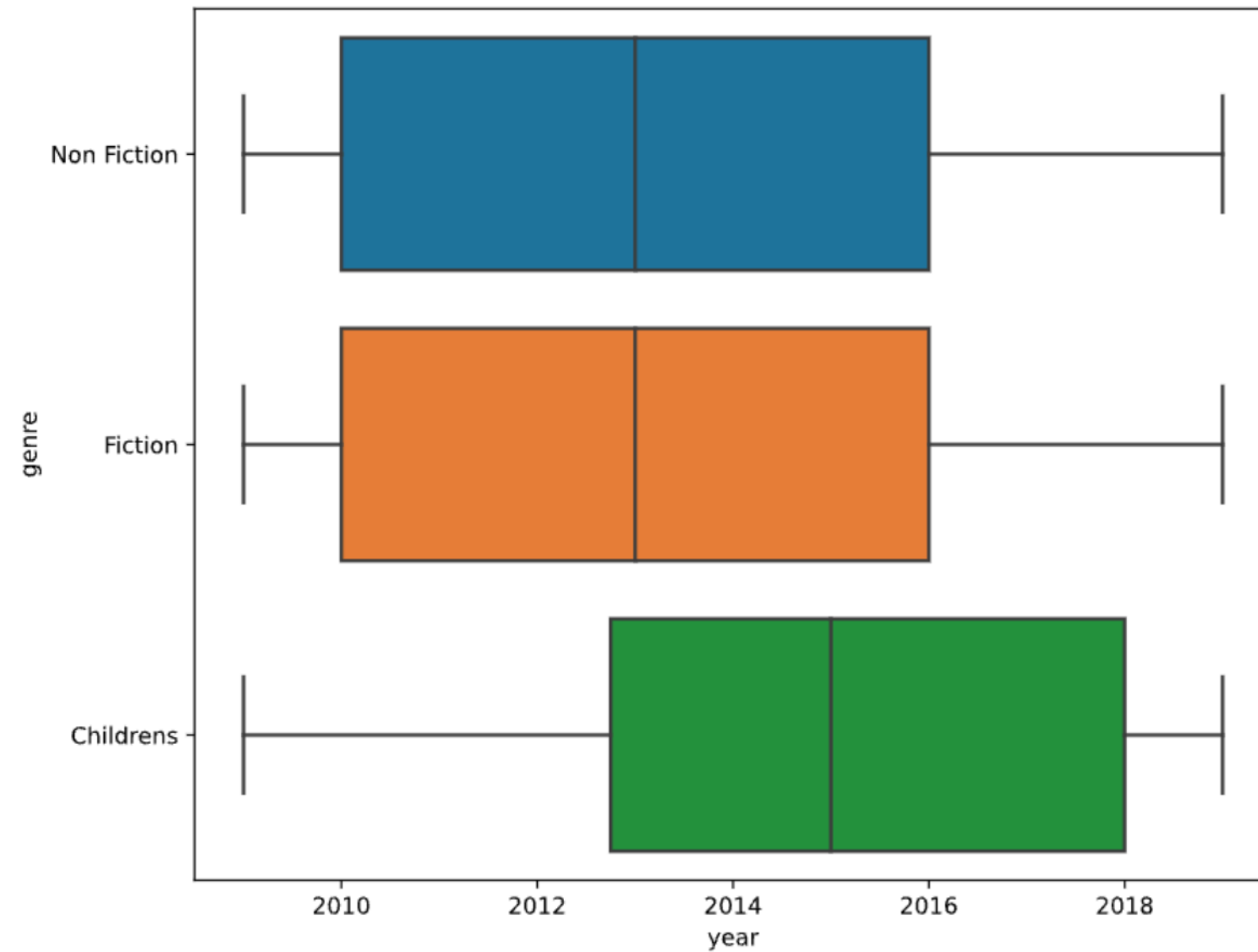
2019

```
sns.boxplot(data=books, x="year")  
plt.show()
```



Validating numerical data

```
sns.boxplot(data=books, x="year", y="genre")
```



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Data summarization

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Exploring groups of data

- `.groupby()` groups data by category
- Aggregating function indicates how to summarize grouped data

```
books[["genre", "rating", "year"]].groupby("genre").mean()
```

```
|      genre | rating |      year |
|-----|-----|-----|
|  Childrens | 4.780000 | 2015.075000 |
|    Fiction | 4.570229 | 2013.022901 |
| Non Fiction | 4.598324 | 2013.513966 |
```

Aggregating functions

- Sum: `.sum()`
- Count: `.count()`
- Minimum: `.min()`
- Maximum: `.max()`
- Variance: `.var()`
- Standard deviation: `.std()`

Aggregating ungrouped data

- `.agg()` applies aggregating functions across a DataFrame

```
books[["rating", "year"]].agg(["mean", "std"])
```

```
|      | rating |      year |  
|-----|-----|-----|  
| mean | 4.608571 | 2013.508571 |  
| std  | 0.226941 |      3.28471 |
```


Specifying aggregations for columns

```
books.agg({"rating": ["mean", "std"], "year": ["median"]})
```

	rating	year
mean	4.608571	NaN
std	0.226941	NaN
median	NaN	2013.0

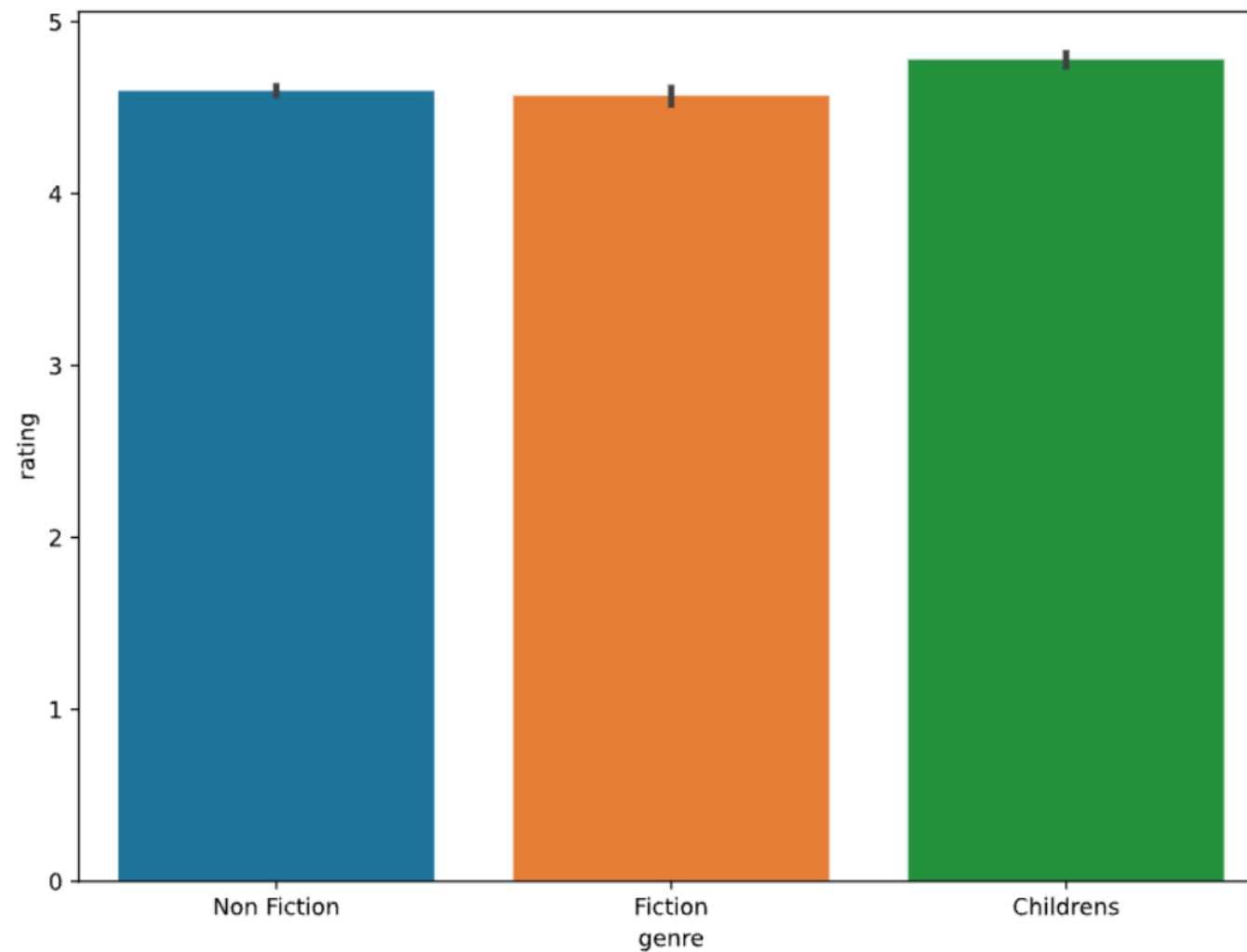
Named summary columns

```
books.groupby("genre").agg(  
    mean_rating=("rating", "mean"),  
    std_rating=("rating", "std"),  
    median_year=("year", "median")  
)
```

genre	mean_rating	std_rating	median_year
Childrens	4.780000	0.122370	2015.0
Fiction	4.570229	0.281123	2013.0
Non Fiction	4.598324	0.179411	2013.0

Visualizing categorical summaries

```
sns.barplot(data=books, x="genre", y="rating")  
plt.show()
```



Let's practice!

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